

NINAD NAIK

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EDUCATION

Bachelor of Engineering in Information Technology

University of Mumbai, Thadomal Shahani Engineering College, Mumbai

2021 - 2025

CGPA: **8.54 / 10**

WORK EXPERIENCE

Backend Developer, Commotion (acq. by Tata Communications) (gocommotion.com)

January 2025 - Present

- Promoted from Backend Developer Intern (January 2025 - June 2025)
- Engineered **low-latency backend architectures** using NestJS and FastAPI, utilizing targeted database indexes and query optimization to handle **high-concurrency enterprise workloads**.
- Developed **MCP frameworks and A2A orchestration patterns** for real-time contextual voice and text AI agents.
- Built robust **AI integrations** with platforms like Salesforce, ServiceNow, and SAP, unifying multi-system workflows.
- Enforced security using OAuth and JWT auth schemes, ensuring platform integrity via SAST and VAPT security analysis.
- Developed **low-latency Voice AI** solutions in collaboration with **multiple Tata companies and MNCs** to streamline automated operations.

Full Stack Developer Intern, Arxena Inc. (arxena.com)

June 2024 - July 2024

- Improved efficiency of the automated recruitment process by developing a **GPT-powered WhatsApp chatbot** in NestJS, **WebSocket API, GraphQL and React.js**, streamlining communication with candidates.
- Optimized build process of the apps running on **AWS EC2** using **AMI**, reducing costs by upto **75%** (detailed in [Gist](#)).

OPEN SOURCE MENTORSHIP

The Linux Foundation - LFX - Linux Kernel Mentorship Program mentee, Spring 2026 ([link](#))

March 2026 - August 2026

- Graduated with **11 patches** [contributed](#) to Linux Kernel mainline; published a [blog](#) sharing my experience.
- Converted text-based **Devicetree bindings** to JSON Schema; fixed documentation links, typos, and driver error handling.
- Worked with **kernel development workflow, patch submission, and review cycles** and used tools like **cscope**, **Cocinelle**, and **smatch** for static analysis. Selected for the mentorship program from **693 applicants**.

PROJECTS

Redspot, FastAPI, React.js, Clickhouse, Kafka, PostgreSQL, Docker | ([GitHub](#))

June 2026

- Built a **real-time website click analytics platform** with a FastAPI ingestion server, Kafka message queue, and ClickHouse storage pipeline that captures user interactions via a lightweight JavaScript tracking script.
- Developed a React dashboard that renders target websites in an iframe and overlays a **color-coded heatmap visualization of click density** using canvas, powered by live data queried from ClickHouse

SpeakSure, Python, Flask, Tensorflow, React.js, MongoDB | ([GitHub](#))

October 2025

- Developed an **Automated AI Behavioural Interview system** that quantifies **confidence level** in candidate's speech and provides AI insights to the hiring manager. Presented **research paper** at ICCIS 25 based on new models and findings.
- Trained **multilayer perceptron (MLP)** and **Fuzzy Hypersphere Neural Network** models on a custom made dataset with **751** audio samples of 30 seconds each by extracting features with Wav2Vec2 and achieved **96%** relaxed testing accuracy.

Shortomega, React.js, Next.js, Node.js, NestJS, TypeScript, Redis, Docker | ([Website](#), [GitHub](#))

November 2024

- Developed a **Full Stack URL Shortening App** that generates short URLs for long URLs, with a focus on system design.
- Published a [technical blog](#) describing the methods of using **Redis as a Primary Database**.
- Implemented **Dashboard, Analytics data aggregation and Rate Limiting** features, and **Dockerized** the application.

SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript, Go

Frameworks: Node.js, NestJS, FastAPI, MCP, Kafka

API & Security: REST, GraphQL, gRPC, OAuth2.0, JWT

Web: HTML, CSS, Tailwind CSS, React.js

Databases: MySQL, PostgreSQL, MongoDB, Redis

Platforms & Tools: Linux, Docker, Kubernetes, AWS

ROLES OF RESPONSIBILITY

Responsible Computing Fellow, SimPPL (simppl.org)

March 2024 - June 2024

App Team Member, Developers' Club - TSEC ([LinkedIn](#))

March 2023 - March 2024

Core Team Member, Google Developer Student Clubs - TSEC ([Website](#))

August 2022 - April 2024

ADDITIONAL INFORMATION

- Presented research paper **Evaluating Human Speech Confidence with Deep Learning Techniques** by Ninad Naik, et al. at **ICCIS 2025, BITS Pilani, Goa**. Accepted for publication in SCOPUS-indexed **Springer LNNS**. ([Certificate](#))
- Open Source Contribution to the [Linux Kernel](#), n8n.io and [CircuitVerse](https://circuitverse.org).
- First Runner Up (CSS Battle 2022 & Javagenix 2022), 2nd Runner Up (PyExpo 2023), Top 10 (ACE Hacks 2024)
- Maintaining a **tech blog** to share experiences, learnings and results on tech.ninadnaik.me